

The symmetry of 2D shapes; congruent shapes

A Cambridge insert: lines of symmetry, order of rotation, and when two shapes are the same.

LESSON 114–115

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 8.1, 8.3

LESSON CLOCK

20'	25'	25'	8'	2
Lines of symmetry	20 min			
Rotational symmetry	25 min			
Congruent shapes	25 min			
In the world	8 min			
Homework	2 min			

LEARNING OBJECTIVES

- Find the lines of symmetry of a shape.
- Find the order of rotational symmetry.
- Decide whether two shapes are congruent.
- Recognise symmetry in ornament and in nature.

TEXTBOOK

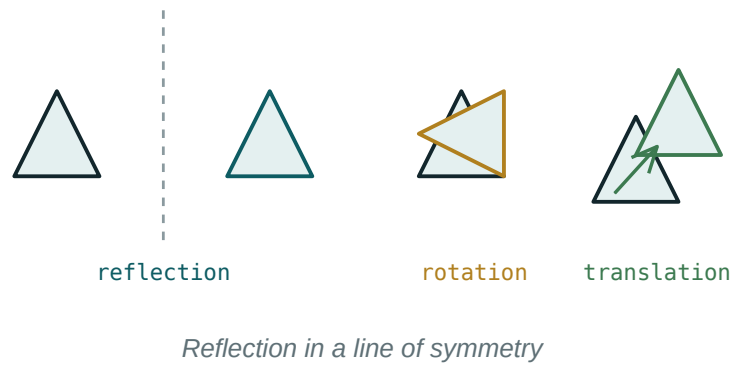
National	pp. 329–334
Cambridge	Stage 7, pp. 78–86
Workbook	Exercise 8.1

01

Explanation

Lines of symmetry

A **line of symmetry** folds the shape onto itself exactly.



SHAPE	LINES OF SYMMETRY
square	4
rectangle (not square)	2
equilateral triangle	3
isosceles triangle	1
scalene triangle	0
rhombus	2
regular hexagon	6
circle	infinitely many

A RECTANGLE HAS TWO LINES, NOT FOUR

Its diagonals do not fold it onto itself — the folded corner does not land on a corner. Testing with a folded piece of paper settles it.

Rotational symmetry

The **order** is the number of positions, in one full turn, where the shape looks the same.

SHAPE	ORDER	ANGLE OF ROTATION
square	4	90°
rectangle	2	180°
equilateral triangle	3	120°
regular hexagon	6	60°
parallelogram	2	180°
isosceles triangle	1	none but the full turn

ORDER 1 MEANS NO ROTATIONAL SYMMETRY

Every shape returns to itself after 360° , so the order is never zero. A regular n -sided polygon has order n and n lines of symmetry.

Congruent shapes

Two shapes are **congruent** when one can be moved exactly onto the other by a translation, a rotation, a reflection, or a combination of them.

PAIR	CONGRUENT?	WHY
two squares of side 5	yes	same shape and size
a square of 5 and one of 6	no	different size
a triangle and its mirror image	yes	reflection is allowed
a shape and its enlargement	no	similar, not congruent

CONGRUENT IS “IDENTICAL”; SIMILAR IS “SAME SHAPE”

Two photographs of different sizes are similar; two copies of the same photograph are congruent. Grade 9 will separate the two ideas properly.

In the world

WHERE	SYMMETRY
a butterfly	one line
a snowflake	six lines, order 6
a wheel with 8 spokes	order 8
an Uzbek ornamental star	often order 8 or 16
the letter <i>H</i>	two lines, order 2
the letter <i>S</i>	no lines, order 2

REFLECTION AND ROTATION ARE INDEPENDENT

S has rotational symmetry but no line; an isosceles triangle has a line but no rotation. Having one says nothing about the other.

02 Worked examples

EXAMPLE 1 How many lines of symmetry has a rectangle that is not a square? What is its order of rotational symmetry?

- ① Two lines: through the midpoints of opposite sides.
- ② The diagonals are not lines of symmetry.
- ③ Rotating 180° leaves it unchanged, so the order is 2.

ANSWER

2 lines, order 2

EXAMPLE 2 A regular octagon: how many lines of symmetry, and what order?

- ① A regular n -gon has n lines and order n .
- ② Here $n = 8$.
- ③ Eight lines, order 8; the rotation angle is 45° .

ANSWER

8 and 8

EXAMPLE 3 Are a triangle and its mirror image congruent?

- ① Congruence allows reflection.
- ② The two have the same sides and the same angles.
- ③ So yes.

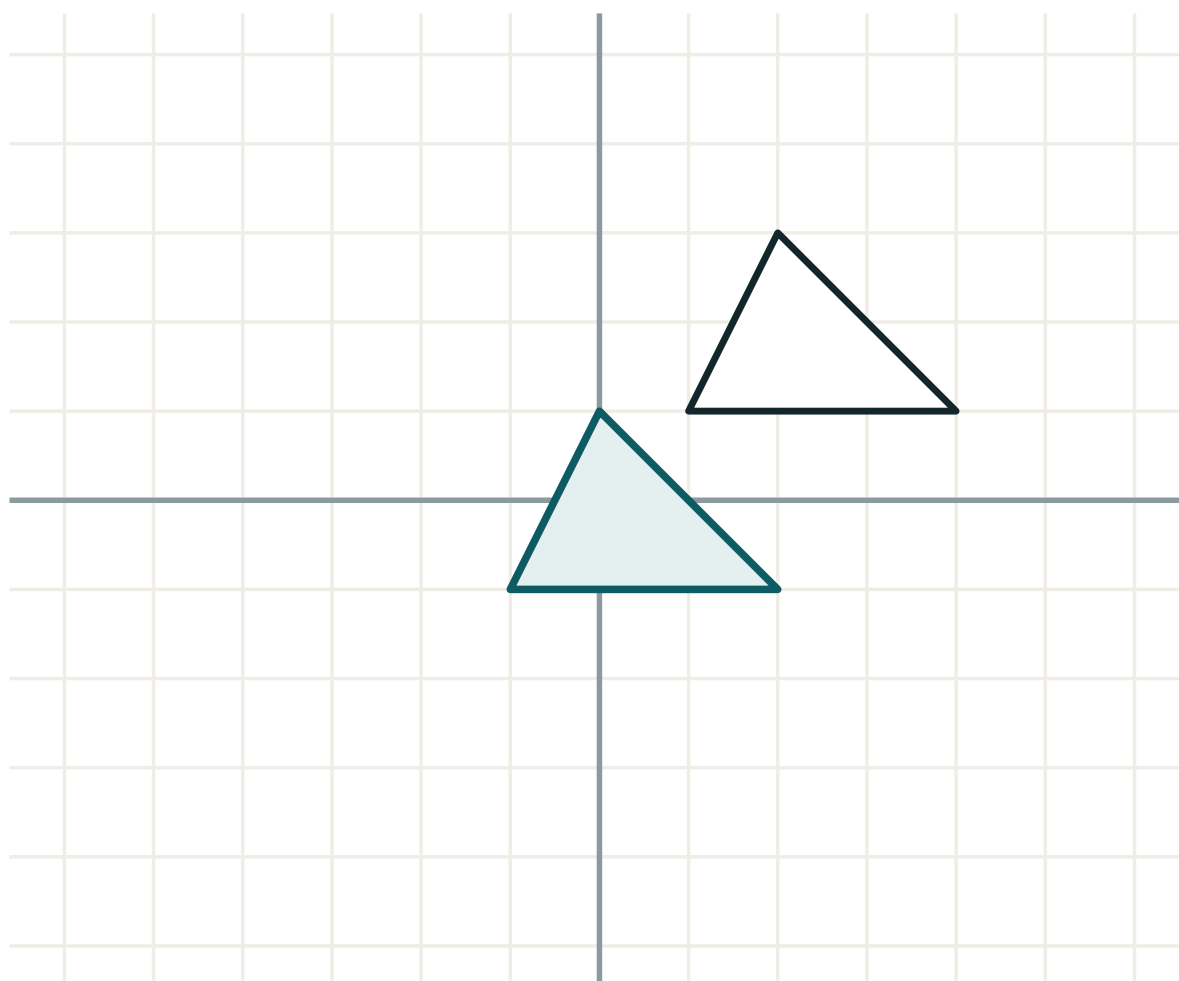
ANSWER

Yes

03 Interactive model

Fold paper shapes to find the lines and turn them on a pin to find the order; both symmetries become physical before they become vocabulary.

Reflect, rotate and translate a shape



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Symmetry	Simmetriya	Симметрия
Line of symmetry	Simmetriya o'qi	Ось симметрии
Rotational symmetry	Aylanma simmetriya	Поворотная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Congruent	Teng (mos)	Равные (конгруэнтные)
Reflection	Simmetrik akslantirish	Отражение
Rotation	Burish	Поворот
Translation	Parallel ko'chirish	Параллельный перенос

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A square has how many lines of symmetry?

A rectangle that is not a square:

An equilateral triangle has order:

The order of rotational symmetry is never:

0

1

2

6

Two shapes are congruent when:

they look alike

one fits exactly on the other

they have the same area

they are both regular

A shape and its enlargement are:

congruent

similar

identical

unrelated

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Lines of symmetry of a square

ANSWER 4

2. Of a rectangle

ANSWER 2

3. Of an equilateral triangle

ANSWER 3

4. Of an isosceles triangle

ANSWER 1

5. Of a scalene triangle

ANSWER 0

6. Of a circle

ANSWER infinitely many

7. Order of a square

ANSWER 4

Medium · the standard question

7 PROBLEMS

1. Order of a rectangle

ANSWER 2

2. Order of an equilateral triangle

ANSWER 3

3. Order of a regular hexagon

ANSWER 6

4. Order of a parallelogram

ANSWER 2

5. Lines of symmetry of a rhombus

ANSWER 2

6. Are two squares of side 5 congruent?

ANSWER Yes

7. Are a shape and its enlargement congruent?

ANSWER No — similar

Hard · stretch and reason

7 PROBLEMS

1. A regular octagon: lines and order

ANSWER 8 and 8

2. A regular n -gon: lines and order

ANSWER n and n

3. A shape with order 2 but no line of symmetry

ANSWER The letter S, or a parallelogram

4. A shape with a line of symmetry but order 1

ANSWER An isosceles triangle

5. The rotation angle for order 5

ANSWER 72°

6. Is a rotation of 360° counted?

ANSWER Yes — it is the last of the order

7. Why is a rectangle's diagonal not a line of symmetry?

ANSWER The folded corner does not land on a corner

07 Homework

Homework — 5 tasks

Fold to find lines; turn to find the order.

- Find the lines of symmetry and the order for a square, a rectangle and a rhombus.
- Find them for an equilateral, an isosceles and a scalene triangle.
- Find the order of rotational symmetry of a regular pentagon and its rotation angle.
- Write three capital letters with a line of symmetry and two with rotational symmetry.

5. Draw two congruent shapes in different positions and say which movement takes one to the other.